BUSINESS BANK · BUSINESS BANKING TRANSFORMATION OFFICE · BB-BLP-BPR

1.4 Process Performance Baseline

Metric dictionary, stage model, analysis specification, dashboard design and illustrative run

Deliverable	1.4 Process performance baseline per journey: volume, cycle time, touch time, wait time, first-time-right, rework, cost per deal
Version	0.5 – metric definitions and analysis specification (Week 3). v1.0 populated from DR-01/02/03/04/08/12/14 and 1.5 at Week 7, endorsed at G1
Source	Data request 0.7; 1.2 maps (stage and loop definitions); 1.3 matrix (deal-type grouping); 1.5 (touch time); DR-16 (cost rates)
Owner	WS3 Data & MI (Data/MI Analyst)
Accepted by	SteerCo at G1 as the reference for benefits; Head of MI & Regulatory Reporting (definitions)
Tests	Hypotheses A1, A3, A4 (servicing over-processing); C1–C4 (rework and wait dominate cycle time)

Status of the numbers in this document. Section 7 is an illustrative run on assumed values to show the SteerCo the shape of what it will endorse. Every figure there is marked illustrative. The measured baseline is produced in Weeks 5–7 from Bank data. Nothing in section 7 is a finding.

1. Purpose and principles

This is the document the business case is measured against. Every KPI target in Phase 2 and every benefit tracked in Phase 4 refers back to a number defined and measured here. Four principles govern it:

1. **Definitions before data.** Each metric has a precise start event, end event, population and exclusion rule, agreed with MI before the extract is analysed, so that the same definition can be re-measured on the replacement platform.
2. **Distributions, not averages.** Means hide the tail, and the tail is where rework lives. Every time metric is reported at P50, P80 and P95 with the count.
3. **Ranges where sources disagree.** Where DR-02 coverage is partial or survey and timestamps diverge, the baseline is a range. The SteerCo endorses the range.
4. **Restated volume base.** ESR Cleanse is excluded (SD-01, 1.3 §5.1). All percentages use live customer deals (26,745) unless stated.

What the baseline is not

It is not an effort measure (that is 1.5, which feeds the touch-time and cost lines here), not a control assessment (1.6), and not a root-cause analysis (1.9). It is the quantified description of how long things take, how often they go wrong, and what they cost.

2. Stage model

Cycle time is measured against a common stage model so that journeys are comparable. The model is derived from the 1.2 maps; the mapping from BLP status codes to stages is agreed with MI on receipt of the DR-02 sample extract (Week 3) and recorded in Appendix A.

Stage	Start event	End event	Applies to	Maps
S0 Pre-deal	Lead / trigger received (where captured) or customer contact	Deal created	NM, SL only where lead data exists; RN from BDRM report date	*-1
S1 Preparation	Deal created	Deal submitted (validation passed; DA determination made)	All	*-2 R1–D2
S2 Decision	Deal submitted	Decision recorded (approved within DA; or Credit approved / declined)	All	*-2 D2–S5
S2a Credit queue	Transferred to Credit	Assessment started (first Credit action)	Deals outside DA	*-2 S4–K1
S2b Credit assessment	Assessment started	Decision recorded	Deals outside DA	*-2 K1–D4
S3 Documentation	Decision approved	Facility letter marked complete (or “ready for fulfilment” where no letter)	All except CS	*-3 R1–S3
S3a Letter out	FAL created	FAL sent	As S3	*-3 S1–S2
S3b Customer	FAL sent	FAL returned	As S3	*-3 S2–C1
S4 Completion	Decision approved	Ready for fulfilment (all three conditions met)	All except CS	*-3 S3–S6
S5 Fulfilment	Ready for fulfilment	Facility fulfilled / BLP updated	Where facility changes	*-3 R5–E1
S6 SBS	Transferred to SBS	Categorised (or returned)	CR	CR-2 S4–S5
E2E Decision	**Deal created**	**Decision recorded**	**All**	**S1 + S2**
E2E Fulfilment	**Deal created**	**Facility fulfilled**	**All except CS**	**S1–S5**
E2E Customer	**Trigger / first contact (where captured)**	**Facility fulfilled**	**Where S0 exists**	**S0–S5**

Clock rules

- Calendar days, not working days: the customer experiences calendar time. Working-day variants reported for Credit queue (S2a) and Fulfilment (S5) where SLAs are internal.
- Loops do not stop the clock. A deferred deal’s S2 runs from first submission to final decision; the loop is counted separately (section 3, R-metrics).
- A deal reset at the 3-month letter point is one deal with a long S3b, not two deals. Re-originated deals (W-01 batch) are linked to the original by customer and product where DR-01 allows, otherwise flagged.
- Withdrawn and declined deals are included in volume and in S1/S2 timing; excluded from S3–S5.

3. Metric dictionary

ID	Metric	Definition	Unit / stat	Cut by	Source	Hypothesis / objective
VOLUME						
V1	Deals created	Count of deals created in period, by deal type; ESR Cleanse reported separately	Count	Type, segment, month	DR-01	Base for all rates
V2	Deals decided	Count reaching a decision (approved / declined / withdrawn) in period	Count; % of V1	Type, segment, outcome	DR-03	O1
V3	Deals outside DA	Share of decided deals transferred to Credit	%	Type, segment	DR-03	A1, A2, E2
V4	Work in progress	Open deals at period end; age distribution	Count; P50/P80 age	Type, stage	DR-02	O1
V5	Fast-track eligible renewals	Share of renewals meeting all draft criteria (limit unchanged; grade within band; covenants compliant; no watch-list; no EPR in 12 m)	%	Segment	DR-02/04/05	A2
CYCLE TIME (calendar days unless stated)						
T1	E2E Decision time	Deal created → decision recorded	P50 / P80 / P95; n	Type, segment, DA vs Credit	DR-02	O1; A1
T2	E2E Fulfilment time	Deal created → facility fulfilled	P50 / P80 / P95; n	Type, segment	DR-02	O1
T3	Stage time	Per stage S1–S6 per table in §2	P50 / P80; n	Type, segment	DR-02	C4
T4	Credit queue time	S2a; working days	P50 / P80	Type, segment	DR-02	C4
T5	Letter turnaround	S3b: sent → returned	P50 / P80; % > 30 d; % > 90 d	Type, segment, method (post / email)	DR-08	C2
T6	Fulfilment time	S5; working days	P50 / P80	Type	DR-02/12	C4
T7	Renewal lateness	Review date → renewal deal created (negative = early)	P50; % late	Segment	DR-01/04	W-11; O1
T8	Distress to categorisation	CAP Referral created → categorised	P50 / P80	—	DR-14	L10
TOUCH AND WAIT						
W1	Touch time per deal	From 1.5 unit times, by stage and role	Minutes	Type	1.5	O2
W2	Wait time per deal	T2 (hours) − W1 (hours)	Hours; % of E2E	Type, stage	DR-02 + 1.5	C4
W3	Flow efficiency	W1 ÷ (W1 + W2)	%	Type	Derived	C4 (expect < 10%)
QUALITY AND REWORK						
R1	First-time-right at Credit	Share of deals transferred to Credit that reach a decision without a deferral	%	Type, segment	DR-02/03	O4; C1
R2	Deferral loops per deal	Count of “deferred – further info” events per deal	Mean; % with ≥ 2	Type, segment	DR-02	C1

R3	Days added by deferral	T1 for deals with ≥ 1 deferral minus T1 for deals with none (same type)	Days (P50)	Type	DR-02	C1
R4	Letter reset rate	Share of issued letters hitting the 3-month unreturned reset	%; count	Type, segment	DR-08	C2
R5	Post-approval re-processing	Share of approved deals returned to pre-credit after approval (excluding R4)	%	Type	DR-02	C3
R6	EPR chains	Share of EPRs preceded by an EPR; share reaching 3	%	Segment	DR-04	A3
R7	Decline and withdrawal rates	Declined ÷ decided; withdrawn ÷ created	%	Type, segment, stage	DR-03	O4; EX-01
R8	Re-origination rate	Deals reset by the 90-day batch job ÷ created	%	Type	DR-09/02	W-01
COST						
K1	Cost per deal	Σ (W1 per step × loaded rate of role)	£ (central, range)	Type	1.5, DR-16	O2
K2	Annual process cost	Σ K1 × V1 by type	£	Type, role	Derived	O2; business case
K3	Cost of rework	K1 for rework steps × loop counts (R2, R4, R5)	£; % of K2	Type	Derived	O4

4. Data model and quality handling

Join structure

Table	Grain	Key
DR-01 deal register	Deal	deal_id
DR-02 stage history	Deal × transition	deal_id, seq
DR-03 decisions	Deal × decision	deal_id, decision_seq
DR-04 variation / EPR detail	Deal	deal_id
DR-05 risk attributes	Deal (snapshot)	deal_id
DR-08 letter events	Letter	deal_id, letter_seq
DR-12 fulfilment / security	Deal	deal_id
DR-14 CAP events	Deal × event	deal_id, event_seq
1.5 unit times	Type × step group × role	type, step
DR-16 rates	Role	role

Derived fields

- `stage` from status code via Appendix A mapping
- `stage_entered`, `stage_exited` per transition; stage duration
- `loop_count_deferral`, `loop_count_decline`, `reset_flag`, `reprocess_flag`
- `da_route`: within DA / Credit
- `fast_track_eligible` from DR-05 criteria
- `segment` from DR-01 officer code → DR-11 team

Quality rules (from 0.7 §4, applied here)

Rule	Threshold	Handling
Deal count reconciles to RFP (30,561 ± 5%)	Hard	Reconcile before analysis
Stage-path completeness	≥ 85% of deals	Report T-metrics on complete subset; state coverage per type
Timestamp sanity	100% after cleanse	Exclude and count
Join rate DR-01 ↔ 02 ↔ 03	≥ 95%	Investigate keys
DR-08 coverage	May be < 50%	If so, T5 and R4 from RM mailbox sample (walk-the-process) and reported as a range
Outliers	> P99 per metric	Retained in distribution; flagged; investigated for data error vs. real

Coverage statement

Every table in v1.0 carries: population, n after exclusions, exclusion reasons, and the period. A metric with coverage below 70% of its population is reported as indicative, not baseline.

Segment and Corporate handling

Segment cuts use the DR-11 team mapping. Corporate deals are analysed by the Bank BA; the partner receives aggregates only (SD-02).

5. Analysis specification (Weeks 4–7)

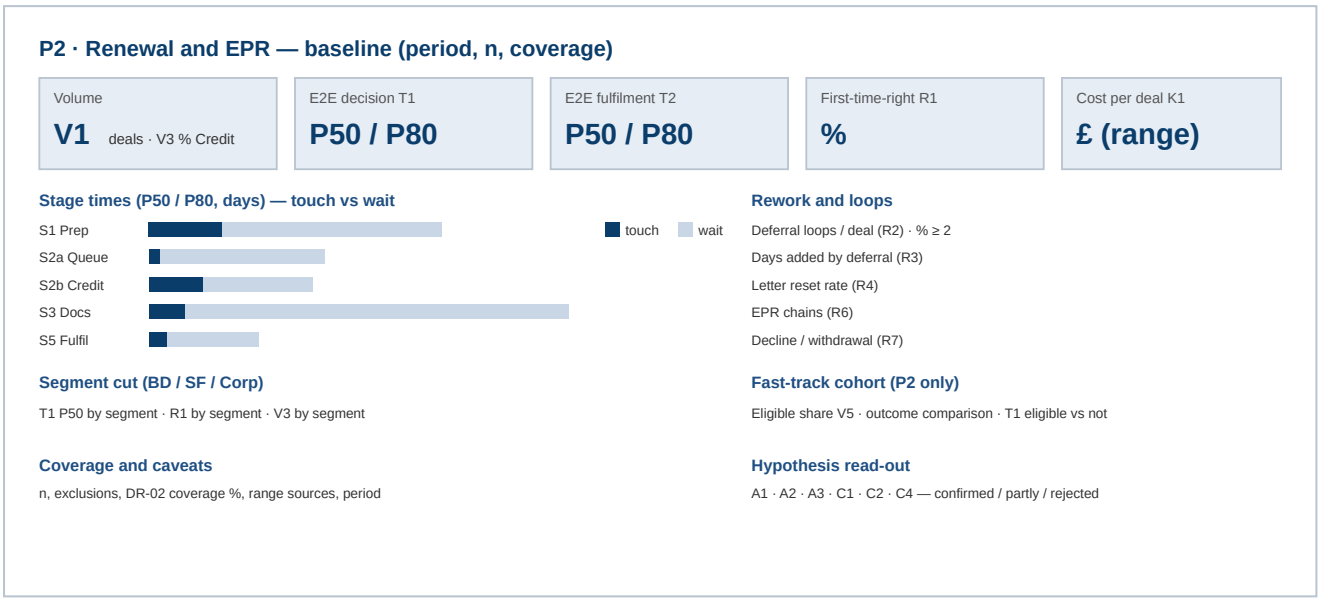
Step	Week	Analysis	Output	Tests
1	3	Sample extract: confirm status-code → stage mapping with MI; agree Appendix A	Appendix A v1	—
2	4	Load, cleanse, reconcile volumes; coverage per deal type	Coverage statement	—
3	4	Stage-path analysis: sequence of stages per deal type; share following the full path	Path table; Sankey per type	A1 (renewal path = new money path?)
4	5	Cycle-time distributions T1–T8 by type and segment; DA vs Credit split	Distribution tables and charts	O1 baseline
5	5	Loop analysis R1–R3, R5, R8: deferral counts, days added, re-processing	Rework tables	C1, C3
6	5	Letter analysis T5, R4 from DR-08 (or mailbox sample)	Letter table	C2
7	5	EPR sequence analysis R6; variation reason classification (DR-04 + sample of 50)	Tables	A3, A4
8	6	Fast-track cohort V5: eligible share; 12-month outcomes (DR-06) vs full-review cohort	Cohort table	A2
9	6	Touch vs wait W1–W3 using 1.5 unit times; flow efficiency per type	Wait tables	C4
10	6	Cost K1–K3 with DR-16	Cost tables	O2
11	7	Assemble baseline pack: one dashboard per journey pattern; ranges; coverage; hypothesis read-out	1.4 v1.0; input to 1.11	All

Hypothesis test criteria (pre-registered)

Hyp.	Confirmed if	Metric
A1	≥ 90% of renewals follow the same stage sequence as New Money	Step 3
A2	≥ 40% of renewals eligible AND eligible cohort's 12-month downgrade / arrears rate ≤ full-review cohort	V5; step 8
A3	≥ 30% of EPRs preceded by an EPR	R6
A4	≤ 30% of variations change limit, security or pricing	Step 7
C1	R1 < 80% (i.e. ≥ 20% of Credit-routed deals defer) AND R3 ≥ 5 days	R1, R3
C2	S3b is the longest single stage at P80 AND R4 ≥ 5%	T3, T5, R4
C3	R5 ≥ 10% on New / Increase	R5
C4	W2 ≥ 3 × W1 at S2a and S5; flow efficiency < 10%	W2, W3

6. Dashboard design

One page per journey pattern (P1–P5 from 1.3), same layout, so the SteerCo reads five pages the same way. The layout below is the template; the illustrative version in §7 shows it filled with assumed numbers for P2 Renewal.



Template. Tiles use the metric IDs from §3 so each number is traceable. Stage bars show touch (dark) and wait (light) at P50; P80 shown as a tick in the populated version.

7. Illustrative run (assumed values — not findings)

Values below are assumptions chosen to be plausible for a UK business bank with a manual facility-letter process and no workflow timers. They exercise the metric definitions and the dashboard layout so the SteerCo can agree the format before Week 7. They are not Bank data and must not be quoted. Replace every yellow cell with measured values.

7.1 Illustrative headline table by pattern

Pattern	V1 deals	V3 % Credit-routed	T1 decision P50 / P80 (d)	T2 fulfilment P50 / P80 (d)	R1 FTR at Credit	R4 letter reset	W3 flow efficiency	K1 £ / deal
P1 New / Increase	3,033	70%	24 / 45	58 / 110	62%	8%	6%	820
P2 Renewal + EPR	14,857	45%	14 / 31	35 / 75	74%	5%	5%	310
P3 Variation (+ removal)	7,277	55%	16 / 36	40 / 85	70%	6%	5%	350
P4 Pre-approved drawdown	62+	15%	5 / 12	18 / 40	90%	3%	8%	220
P5 Distressed	1,516	100%	21 / 48 (to categorisation T8)	n/a	80%	n/a	n/a	450

7.2 Illustrative stage breakdown: P2 Renewal (P50 days; touch from 1.5 §6.2 assumptions)

Stage	P50 days	P80 days	Touch (h)	Wait (d, approx.)	Reads as
S1 Preparation	6	14	2.3	5.7	RM gathers information; customer wait inside
S2a Credit queue (Credit-routed only)	4	9	0	4.0	Pure wait
S2b Credit assessment	4	10	1.0	3.9	Includes deferral loops
S3 Documentation (letter out → returned → checked)	18	42	0.6	17.9	Longest stage; customer-side; 3-month reset tail
S5 Fulfilment	3	7	0.25	2.9	Queue via message
E2E fulfilment (T2)	**35**	**75**	**4.2**	**34.4**	**Flow efficiency ≈ 4.2 h ÷ 840 h = 0.5% of elapsed; ≈ 5% of working hours**

7.3 Illustrative rework table: P2 Renewal

Metric	Value	Reads as
R1 First-time-right at Credit	74%	26% of Credit-routed renewals defer at least once
R2 Deferral loops per deferred deal	1.4 (18% with ≥ 2)	A tail of multi-loop deals
R3 Days added by deferral (P50)	+9	Would meet C1 threshold (≥ 5)
R4 Letter reset rate	5%	c. 740 renewals a year back to pre-credit for a timing failure
R6 EPR preceded by EPR	35%	Would meet A3 threshold (≥ 30)
V5 Fast-track eligible	48%	Would meet A2 volume threshold; outcome comparison still required

What the illustration shows about the method: if the measured renewal profile is anything like this, three facts would drive the design: wait is more than 90% of elapsed time, the documentation stage is half of it, and a

quarter of Credit-routed renewals loop at least once. The levers would then be letter automation (L4), fast-track routing (L1) and submission validation (L11), in that order of cycle-time impact. The measured run in Week 7 will confirm, re-order, or overturn that.

7.4 What v1.0 will contain

- §7.1 table measured, with ranges and coverage per cell
- Five dashboards (P1–P5) per the §6 template, plus segment variants for P2 and P3
- Stage-path Sankeys per deal type; cycle-time distributions (histograms) for T1 and T2 per pattern
- Fast-track cohort analysis (V5 and outcomes) in full
- Hypothesis read-out A1–A4, C1–C4 against the pre-registered criteria in §5
- Appendix A: status-code to stage mapping agreed with MI; Appendix B: exclusions and data-quality log

Appendix A. Status-code to stage mapping (agreed with MI on sample extract, Week 3)

BLP status / workflow code	Description	Stage (§2)	Start or end event for	Notes
[DR-02 code]	Deal created	S1 start	T1, T2	
[DR-02 code]	Validation passed / submitted	S1 end; S2 start	T3	
[DR-02 code]	Transferred to Credit	S2a start	T4	V3 flag
[DR-02 code]	Credit assessment started	S2a end; S2b start	T4	If not captured, S2a + S2b reported combined
[DR-02 code]	Deferred – further information	loop event	R1, R2, R3	
[DR-02 code]	Approved (DA) / Approved (Credit) / Declined / Withdrawn	S2 end	T1; R7	DR-03 authority level
[DR-02 code]	FAL created / sent / returned / complete	S3a, S3b	T5, R4	DR-08; coverage TBC
[DR-02 code]	Pre-conditions met; security complete	S4	T3	
[DR-02 code]	Ready for fulfilment	S4 end; S5 start	T6	
[DR-02 code]	Fulfilled / BLP updated	S5 end	T2	
[DR-02 code]	Transferred to SBS; Categorised	S6	T8	DR-14
[DR-02 code]	Re-originated (batch)	reset event	R8	Link to original deal